

After hours teleworking and boundary management: Effects on work-family conflict

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Abstract.

BACKGROUND: There is a need for a better understanding of how the use of technology to complete work outside regular office hours is related to work-life outcomes. Few studies have also investigated how individual differences in work-nonwork boundary management relate to work-life outcomes.

OBJECTIVE: This study was conducted to examine how teleworking outside regular office hours and individual boundary management relate to work-family conflict.

METHODS: A web survey was sent to fulltime employees at the headquarters of a multinational high-tech firm in Sweden. A total of 71 answers were obtained and analyzed using regression analysis.

RESULTS: The extent of teleworking after hours was unrelated to work-family conflict. However, as previous research has shown, having more permeable boundaries and allowing work to interrupt nonwork behavior was related to higher levels of conflict.

CONCLUSIONS: The findings suggest that teleworking after hours is not as problematic in terms of work-family conflict as has been reported in previous studies. Furthermore, in order to prevent high levels of work-family conflict, it is seemingly beneficial to avoid work interruptions during nonwork behavior.

Keywords: Boundary permeability, role conflict, telecommuting, work-nonwork boundaries

1. Introduction

Teleworking (or telecommuting) typically involves using computer technology to work away from the main worksite for a proportion of the workweek [1]. It offers many potential benefits for employees and employers alike [2, 3]. One often-stressed benefit for employees is increased flexibility. The increased

flexibility may help employees to better regulate and synchronize the demands between work and family [4]. As a result, teleworking has popularly been presented as a way to alleviate work-family conflict - a conflict stemming from competing roles in the work and family domains [5]. However, evidence has been inconsistent [4, 6–8].

Previous research on work-family conflict has generally studied teleworking during regular office hours [9], with some exceptions [9–13]. Studies on the effects of teleworking outside regular office hours are thus needed. It is warranted given our limited understanding of how the use of technology to complete

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work outside the regular office hours relates to important aspects of work and work-life outcomes [14].

Furthermore, as teleworking from home alters the traditional boundaries (e.g. physical, psychological, temporal) between work and family domains [15], work-family conflict could increase unless the boundaries are managed effectively. Yet, how employees manage the boundaries between work and nonwork has until more recently been given little attention [16]. In fact, little is known about how differences in management of such boundaries relate to work-life outcomes [17].

This study addressed both these research gaps by examining how teleworking (from home) after hours and boundary management affect work-family conflict. First, it examined how the extent of teleworking after regular office hours is related to both directions of the work-family conflict, i.e. work interference with family and family interference with work. To enable comparisons, both telework during and outside regular office hours were measured and included in the analysis. However, focus was on the latter – defined in this study as voluntary or involuntary work conducted from home with the aid of information and communications technology outside a regular Monday to Friday 8 am to 5 pm workweek. The study also examined how differences in boundary management relate to work interference with family and family interference with work. Boundary management was studied as the degree of boundary permeability a person allows between work and nonwork roles.

2. Theory and hypotheses

2.1. *Work-family conflict*

Work-family conflict (WFC) can be defined as ‘a form of inter-role conflict in which the role pressures from the work and family domains are mutually incompatible in some respect’ [18]. The role pressure is thought to generate moods or states, which affect wellbeing in both the work and family domain [19]. Indeed, WFC has been linked to adverse outcomes such as psychological distress, depression, somatic complaints, and exhaustion [5]. It has been found to influence e.g. job satisfaction, life satisfaction, organizational commitment, and turnover [20]. Of little surprise, ways to reduce and prevent WFC is of considerable interest for employers and managers of organizations.

WFC has been found to have two distinct dimensions: work interference with family (WIF) and family interference with work (FIW) [7, 20–23]. WIF occurs e.g. when one has to skip family activities in order to be able to complete work tasks. FIW occurs e.g. when family responsibilities interfere with one’s work responsibilities. Although both types of conflict have been found to be consistently negatively related to desirable outcomes, the negative effect of WIF has been found to be greater [12]. That is, WIF is more problematic than FIW.

Researchers have looked at different forms of conflict depending on the source [22]. The emphasis in research has mostly been on time-based and strain-based conflict [7, 9]. Time-based conflict occurs when the amount of time devoted in one domain (work or family) interferes with performing responsibilities in the other domain [23]. Strain-based conflict occurs when strain created in one domain interferes with performing responsibilities in the other. Both forms were measured in this study. However, no distinction was made in the presentation of the results as they tend to be intertwined [7].

2.2. *Telework after hours and work-family conflict*

Researchers have become increasingly aware of the benefit of looking at the extent of teleworking in order to understand its impact on work-family conflict [9]. However, in work-family conflict research, telework has generally been studied as something that may only take place during regular office hours. This is understandable given that it is usually perceived as a form of work where hours at the office are substituted with hours at home [9, 24]. Nonetheless, teleworking is also conducted outside regular office hours, i.e. “after hours”. For employees who have flextime and therefore may alter workday start and finish times (as in this study), this type of work does not necessarily have to represent supplemental work or overtime in contrast to “traditional” office employees (who may only telework after their workday at the office or not at all).

Telework, and telework from home after hours more specifically, has both potential costs and benefits for employees. Potential benefits that may lower WFC include e.g. increased flexibility, and increased control over the pacing and scheduling of the work, as well as opportunity to spend more time at home with one’s family [10, 11]. However, we believe teleworking outside regular office hours is generally negative

for both types of WFC. For instance, the physical demarcation between work and family domains is erased for employees when they work from home [25]. This could lead to more distractions and interruptions from work when a person is in a family role (and vice versa), increasing levels of WIF and FIW. A lack of demarcation between roles at home may also make it hard to mentally disengage from work [9]. In addition, when an employee is teleworking from home after hours, he or she is likely detracting from personal/family time since work is conducted outside regular office hours. It might also be used as a form of supplemental work and augment the amount of hours worked per week. This is problematic because studies have linked work hours to increased levels of conflict [26].

Previous studies on teleworking outside regular office hours, treating it as a form of supplemental work, have found that it is linked to higher WIF. For instance, Boswell and Olson-Buchanan [14] as well as Fenner and Renn [13] found conducting work from home using technology after regular office hours was related to higher WIF. Duxbury et al. [10] similarly found a positive relationship between teleworking after hours and WIF. Duxbury et al. [11] found that employees teleworking after hours had higher levels of WFC (not specifically WIF though) than those not engaged in this form of work. Hence, we propose:

H1. The extent of teleworking outside regular office hours is related to higher work interference with family (WIF).

Duxbury et al. [10] investigated the relationship between teleworking outside regular office hours and FIW. No significant relationship was found. Yet, as teleworkers weaken the boundaries that normally separate the work domain from the family domain, they may become more accessible for demands of family members [7, 27]. This might lead to frequent, unwanted work interruptions, and consequently increased FIW. Other family members might for example not acknowledge or respect that work is conducted during evenings, early mornings and weekends. This in combination with risks such as role-confusion and problems with role-detachment (e.g. from a family role) [28] should lead to an increase rather than decrease in FIW. Hence, we propose:

H2. The extent of teleworking outside regular office hours is related to higher family interference with work (FIW).

2.3. Boundary management and work-family conflict

There is currently little knowledge how work-nonwork boundary management relates to work-family conflict, in part because researchers only recently have begun to thoroughly examine such boundaries and how they are managed [16].

Two related and highly similar theories, but with different research origins, deal with the management of boundaries [29]. These are boundary theory [28, 30] (which focuses on work and *life* domains), and work-family border theory [31] (which focuses more on work and *family* domains). According to these theories, individuals actively and deliberately create boundaries between work and personal/family life roles. Some create boundaries to segment roles and domains, while others maintain weaker boundaries to integrate the roles. The theories suggest these preferences range on a continuum from segmentation to integration, but the actual level of integration/segmentation is influenced by factors such as family, co-workers, and type of occupation.

In the integration or segmentation of roles/domains, the strength of boundaries plays a central role. It is affected by flexibility and permeability [28, 31]. Flexibility is the extent to which a boundary may contract or expand (in time and space) depending on the demands in one domain or the other [31]. For example, a work domain boundary has a high degree of flexibility (in terms of time and space) if the employee can choose both when and where to work. The permeability of a boundary is the degree to which elements from other domains may enter [31]. Such permeations are often perceived as interruptions. When boundaries have a high degree of flexibility and permeability, there is more integration between work and nonwork roles/domains.

The theories suggest there are both costs and benefits with integrating and segmenting, respectively [28, 31]. An important advantage of integration is the possibility of easier, less demanding transitions between work and family roles/domains [28]. However, integration can create increased complexity as work and nonwork roles/domains start to blur [28, 31]. More specifically, higher integration may lead to role-confusion (i.e. problems with determining whether a work or family role should be enacted), role-attachment issues (i.e. problems with switching between work and family/life when needed), and more frequent cross-boundary interruptions [28].

Previous empirical research has also found that higher integration (in the form of higher permeability between work and nonwork) is linked to higher rather than lower WFC. Some studies have reported a positive relationship between WIF and work to nonwork permeability, [16, 32, 33]. Some studies have reported a positive relationship between FIW and nonwork to work permeability [16, 32]. Moreover, Kossek et al. [17] found that higher integration was linked to an increase in FIW. Kim and Hollensbe [34] reported a positive relationship between *home-to-work* conflict (a similar form of conflict) and nonwork to work permeability. Hence, we expect the following:

H3. Higher work to nonwork permeability is related to higher work interference with family (WIF).

H4. Higher nonwork to work permeability is related to higher family interference with work (FIW).

3. Method

3.1. Study population

The sample consisted of fulltime employees from two sales departments at the Swedish headquarters of a multinational high-tech firm with over 100,000 employees worldwide. The employees, who were all able to telework (e.g. from home) some proportion of their workweek, were asked to complete an anonymous web survey. The survey took place during May and June 2014. Reminders were sent by e-mail to those who had not responded.

3.2. Measurements

3.2.1. Work-family conflict

An 8-item measure by Gutek et al. [21] was used to assess WIF and FIW. Although this measure is a WFC scale, it relates to an individual's personal life more generally [14]. The items are relevant even for persons without a family or a partner. The Likert scales ranged from 1 (strongly disagree) to 5 (strongly agree). Items measuring WIF included e.g.: "After work, I come home too tired to do some of the things I'd like to do". Items measuring FIW included e.g.: "I'm often too tired at work because of the things I have to do at home". Internal consistency was acceptable with $\alpha = 0.83$ for WIF and $\alpha = 0.74$ for FIW.

3.2.2. Boundary management/permeability

Items were selected from the permeability dimension of the Work-life indicator by Kossek et al. [32]. These items measure to what extent a person allows work to interrupt nonwork behavior and the opposite. If more cross-boundary interruptions are allowed then the person has more permeable boundaries between work and nonwork. Conversely, if little interruption is allowed then the person has less permeable boundaries. However, a person might allow interruptions only in one direction, which makes it important to acknowledge the direction. Of the selected items, five measured to what extent work to nonwork interruption was allowed, and four allowance of interruptions in the opposite direction. Likert scales were used, ranging from 1 (strongly disagree) to 5 (strongly agree). Internal consistency was acceptable for work to nonwork interruption ($\alpha = 0.78$) and nonwork to work interruption ($\alpha = 0.78$).

3.2.3. Extent of teleworking

Employees were asked: "How many hours do you work from home per week during regular office hours (Mon-Fri, 8 a.m. – 5 p.m.)?", and "How many hours do you work from home per week outside the regular office hours, e.g. evenings and weekends?" "Teleworking" was not mentioned specifically in these questions, and the work measured may therefore encompass work that is not technology-aided as well. However, this must be considered unlikely since employees at the company of interest are dependent on information and telecommunications equipment as a means for work. Moreover, the company's headquarter is an open, activity based workplace without private, reserved offices; therefore storing all work-related material digitally is common practice.

3.2.4. Control variables

Age, gender, number of children living at home, marital status, and work hours per week were included as control variables in all analyses since these consistently have been found to influence levels of WFC [22].

3.3. Statistical analysis

Pearson product-moment correlation coefficients were used in the initial analysis of variables, and hierarchical multiple regression was used for testing the hypotheses. A square root data transformation was used before the regression analyses since "telework during regular office hours" was positively skewed

[35, 36]. Multicollinearity was not a problem since VIF was low (not above 1.9 in any of the analyses) and there were no extreme bivariate correlations [35].

4. Results

4.1. Participant demographics

When the data collection ended, 71 useful responses had been received (representing a 68% response rate). Of the respondents, 65% were men and 35% female, 86% were married or cohabitant, and 89% were parents with children living at home. Most had one (24%) or two children (47%) at home. The respondents held a variety of positions within sales: 68% were salesmen, 14% sales managers or sales directors, 13% had technical sales roles, 4% were business/industry developers, and 1% had administrative roles. As can be seen in Table 1 (which presents means, standard deviations, and correlations) respondents were on average 43 years of age (ranging from 27 to 57), had been employed by the company 8.9 years (ranging from 1 to 23), and worked 48.7 hours per week (ranging from 40 to 60). Teleworking outside office hours amounted to 9.4 hours per week on average (ranging from 1 to 20) and teleworking during office hours to 9.6 hours (ranging from 0 to 30).

4.2. Hypotheses

H1 stated extent of teleworking outside regular office hours would be positively related to WIF. However, as can be seen from Table 2, no significant relationship was found between extent of teleworking outside regular office hours and WIF ($\beta = 0.05$, $p = 0.77$).

H2 stated extent of teleworking outside regular office hours would be positively related to FIW. This was not supported ($\beta = 0.08$, $p = 0.60$). Furthermore, as can be seen in Table 2, extent of teleworking during office hours was not related to WIF ($\beta = 0.25$, $p = 0.05$) or FIW ($\beta = -0.03$, $p = 0.83$). However, there was a tendency towards a significant relationship between extent of telework during office hours and WIF. This makes the result less clear-cut.

H3 stated a higher work to nonwork permeability would be related to higher levels of WIF. This hypothesis was supported. Work to nonwork permeability was related to higher levels of WIF ($\beta = 0.27$,

Table 1
Descriptive statistics and correlations

Variable	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Age	43.03	6.20	—													
2. Gender	1.65	0.48	-0.11	—												
3. Marital status	1.14	0.35	-0.19	-0.04	—											
4. Children	1.77	1.04	0.22	0.21	-0.42**	—										
5. Workhours	48.71	5.78	0.10	-0.06	-0.10	0.06	—									
6. Tenure	8.89	5.56	0.24*	0.15	-0.07	0.06	0.16	—								
7. Occupation	1.38	0.49	-0.01	-0.12	-0.11	-0.11	0.13	0.18	—							
8. Work interference with family (WIF)	2.91	0.96	-0.02	0.14	0.07	-0.00	0.13	0.10	-0.02	—						
9. Family interference with work (FIW)	1.49	0.56	-0.09	0.35**	-0.12	-0.01	-0.04	0.01	-0.09	0.21	—					
10. Extent of teleworking during office hours	9.60	5.64	0.07	0.22	0.18	0.19	-0.02	0.04	-0.14	0.19	0.05	—				
11. Extent of teleworking outside office hours	9.37	4.37	0.14	-0.05	-0.17	0.22	0.62***	0.22	0.22	0.16	0.00	0.03	—			
12. Total extent of teleworking	18.97	7.22	0.14	0.15	0.04	0.28*	0.36**	0.17	0.02	0.25*	0.04	0.80***	0.63***	—		
13. Boundary permeability (work to nonwork)	3.65	0.88	0.09	-0.17	-0.18	0.14	0.17	-0.08	-0.19	0.14	0.00	-0.01	0.32**	0.19	—	
14. Boundary permeability (nonwork to work)	3.35	0.85	-0.14	0.02	-0.22	0.15	-0.26*	-0.18	-0.12	-0.32**	0.24*	0.16	-0.17	0.03	0.21	—

Note. $n = 71$, but $n = 69$ for the 1st variable and 70 for 4th variable. Gender: female = 1, male = 2. Occupation: 1 = salesman, 2 = other. Marital status: 1 = married or cohabitant, 2 = other. Tenure is in years at the company. WIF/FIW: 1 = lowest, 5 = highest. Boundary permeability: 1 = lowest, 5 = highest. Teleworking is in hours per week. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (Two-tailed).

Table 2

Hierarchical regression analyses with work interference with family (WIF) and family interference with work (FIW) as dependent variables

	WIF					FIW				
	β	R^2	Adjusted R	ΔR^2	Final F	β	R^2	Adjusted R ²	ΔR^2	Final F
Step 1		0.05	-0.03				0.16	0.09		
Age	-0.12					0.00				
Gender	0.12					0.39**				
Marital status	-0.05					-0.11				
Children living at home	-0.04					-0.19				
Hours worked per week	-0.05					0.01				
Step 2		0.08	-0.03	0.03			0.16	0.06	0.00	
Extent of teleworking during regular office hours	0.25					-0.03				
Extent of teleworking outside regular office hours	0.05					0.08				
Step 3		0.24*	0.13*	0.16**	2.10*		0.21	0.09	0.05	1.75
Boundary permeability (work to nonwork)	0.27*					-0.01				
Boundary permeability (nonwork to work)	-0.45**					0.26				

Note. $n = 71$, but $n = 69$ for "age" and "children living at home". Values represent standardized regression coefficients (β) from the final equation. * $p < 0.05$, ** $p < 0.01$.

$p < 0.05$). Interestingly, nonwork to work permeability was related to lower levels of WIF ($\beta = -0.45$, $p < 0.005$). Having permeable boundaries seems to have beneficial impact on WIF, but only when nonwork is allowed to interrupt work.

H4 stated a higher nonwork to work permeability would be related to higher levels of FIW. This hypothesis was not supported ($\beta = 0.26$, $p = 0.07$) (Table 2, step 3). When comparing the two analyses in Table 2, some important differences can be seen. In the final model with WIF as dependent variable, $F(9, 59) = 2.10$, $p < 0.05$, 24% of the variance was explained. Among the included variables, boundary permeability (work to nonwork and vice versa) were the only variables that made significant contributions to WIF, R square change = 0.16, $p < 0.01$.

The final model with FIW as dependent variable was not significant, $F(9, 59) = 1.75$, $p = 0.10$, R square change 0.05, F change = 1.87, $p = 0.16$. In fact, the only variable that was significant in this model was gender, $\beta = 0.39$, $p < 0.01$. More precisely, male employees experienced higher levels of FIW than their female colleagues.

5. Discussion

5.1. General discussion

This study was conducted in order to examine how teleworking after hours is related to work interference with family (WIF) and family interference with work (FIW). The purpose was also to examine how boundary management, or more precisely differences in boundary permeability, is related

to WIF and FIW. Data was collected from full-time employees at a multinational high-tech firm. Most of them were male, married or cohabitant, parents with children at home, and working as salesmen.

The results gave no support for hypothesis 1 and 2, which stated the extent of teleworking after hours would be related to higher levels of WIF and FIW. This is contrary to prior studies [10, 13, 14]. Why findings are different from these studies is difficult to pinpoint, but there are several possible explanations. First, Fenner and Renn [13] did not measure extent of telework as in this study. Instead, they used a Likert scale. Second, Boswell and Olson-Buchanan [14] and Duxbury et al. [10] focused on "traditional" office employees and not people who have the ability to telework during regular office hours. All teleworking after hours therefore represented supplemental work (as intended). This was also the case in the study by Fenner & Renn [13]. However, teleworking after hours did not always have to be supplemental work in this study because employees had flextime and could alter the start and end of their workday. Third, skills in goal setting and prioritizing have been found to dampen the adverse effect of teleworking on WIF [13], and it is possible that the employees had such skills. Fourth, research has shown that job control is beneficial in terms of work-family conflict [17, 37]. Employees in this study may have felt they were in control over their work situation (in contrast to being forced by workload and pressures from the employer to work more).

Hypothesis 3, stating higher work to nonwork permeability would be related to higher levels of WIF, was supported. In other words, allowing work to

interrupt nonwork behavior is generally problematic in terms of WIF. Prior research on the relationship between WIF and this form of permeability has reached the same conclusion [16, 32, 33]. It is in line with previous research claiming that a higher integration is related to higher WIF.

Hypothesis 4, stating higher nonwork to work permeability would be related to higher levels of FIW, was not supported but a tendency was found ($p=0.07$). In effect, to attend to family issues when working does not seem to increase family interference with work (FIW), which is surprising. Mainly since the direction of the interruption coincides with the direction of the interference (as with work to non-work permeability and WIF). This result contradicts recent findings [16, 17, 32–34].

However, results showed that permeability (non-work to work) was related to less WIF. It has been suggested that it is important to focus on the direction of the permeability [32], and this finding supports the notion. No prediction was made for this relationship but, as Olson-Buchanan and Boswell [33] claim, it nevertheless makes conceptual sense. Dealing with family/life matters while working should arguably lower work interference with family (WIF). Moreover, interruptions are not always perceived as something negative because they can be positive reminders of one's role in another domain, e.g. the parental role [31]. Nonwork interruptions could be perceived as welcomed work breaks or just a positive change of activity.

Finally, the results showed men tended to have higher levels of FIW than women. Male employees perceived the family to interfere more with their work than their female colleagues. The finding and its magnitude were clear, but nonetheless surprising given the small sample size.

5.2. *Practical implications*

The findings suggest teleworking after regular office hours is not as problematic as has been found in previous telework research. The study also shows employees who allow work interruptions to a greater extent while not working tend to have higher WIF. As this finding suggests, employees should benefit from enacting boundaries between work and nonwork to protect specifically from work encroachments. For instance, working during one's vacation, bringing work materials to family activities, and allowing work to interrupt while with friends or family is generally not recommended to create less

interference from work. There are individual differences when it comes to boundary preferences (degree of integration/segmentation) and how boundaries are handled [16, 22]. Accordingly, in terms of work-family conflict, some may actually benefit from having permeable boundaries. It could be the case for some employees in this study (although not visible at the aggregate level.) Furthermore, because the permeability may be affected by the organizational climate and policies [28, 30, 31], it is important that managers consider organizational signals. Employees might have little control over boundaries as a result of organizational decisions [22, 38]. For instance, simply supplying communication devices has been shown to create expectations among employees to be continuously available [39], i.e. having a more permeable boundary that allows work interruptions.

5.3. *Limitations and directions for further research*

The study was cross-sectional and causality, therefore, cannot be determined. Common method variance could be a problem because the independent variable(s) and the dependent variable(s) were measured at the same time using the same method on the same respondents [40]. The study was conducted on a specific group of individuals and the sample size was small. More studies, using a larger sample, are needed to validate the findings for other types of employees and organizations. It should be noted some relevant variables were not studied, e.g. perceptions of job control. This variable has been found to have a beneficial effect on work-family conflict [17], and it is possible higher levels of job control among the employees could explain the absence of a relationship with WFC. It is also possible a lack of control could explain why more permeable boundaries relate to higher levels of WIF [31]. Therefore future studies should include boundary control as a variable. It is important not the least since an effective work-life strategy will vary depending on individuals' perceptions of boundary control along with factors such as identities and behaviors [41]. Accordingly, research should benefit from studies that fully capture and categorize the individual differences that exist when it comes to managing work-nonwork boundaries, e.g., by using cluster analysis [see e.g., 16, 32]. Lastly, empirical studies referred to in this study, both on boundary management and teleworking after hours, were cross-sectional. The research

within this field would benefit from longitudinal designs to determine direction and causality [see e.g., 25, 29].

6. Conclusions

This study has shown teleworking from home after hours may not be as problematic in terms of work-family conflict as has been shown in some previous studies. A plausible explanation is the sample consisted of professional teleworkers, who are able to choose more freely both when and where to work, and not “traditional” fulltime office employees, i.e. employees restricted to teleworking from home only outside regular office hours (or not at all). Moreover, employees in this study had flextime, which means that the teleworking done after hours did not necessarily have to represent supplemental work.

The study has also shown, in line with previous empirical studies, that higher boundary permeability, and more specifically allowing work to interrupt nonwork roles and behavior, is related to higher work interference with family. Accordingly, employees should benefit from enacting boundaries between work and nonwork to protect from work encroachments (work calls, reminders of work, etc.). To what extent the results in this study can be generalized to other organizations is uncertain. More research is clearly needed to untangle how telework outside regular office hours and individual boundary management impact work-life outcomes such as work-family conflict.

Conflict of interest

None to report.

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